

Predicting Waves from Wind

The SPM (1984) JONSWAP fetch- and duration-limited growth relations for deep-water wind waves, with worked example. Companion note to the online Wind-Wave Prediction calculator.

1. Fetch, duration and fully-developed seas

Waves grow as a wind of speed U transfers energy to the sea over a distance (the **fetch** F) and time (the **duration** t). Three states are possible. A **fetch-limited** sea has had enough time for the waves to grow to the maximum the fetch allows. A **duration-limited** sea is cut short in time: the wind has not blown long enough to develop the full fetch, so the waves are smaller. A **fully-developed** sea is the ceiling for a given wind speed — additional fetch or duration produces no further growth. The calculator evaluates all three and reports the governing (smallest) wave.

2. Adjusted wind speed

Wave growth scales with the wind stress rather than the wind speed itself, so the SPM introduces the wind-stress factor (adjusted wind speed) from the 10-m wind U :

$$U_A = 0.71 \cdot U^{1.23} \quad (U \text{ in m/s}), \quad \text{and the dimensionless fetch is } X = gF/U_A^2.$$

3. Fetch-limited growth (SPM 1984)

For a fetch-limited, deep-water sea the JONSWAP relations give the significant height, peak period and the minimum wind duration needed to reach this state:

$$gH_{m0}/U_A^2 = 1.6 \times 10^{-3} \cdot X^{1/2}, \quad gT_p/U_A = 0.2857 \cdot X^{1/3}, \quad gt/U_A = 68.8 \cdot X^{2/3}.$$

If the actual wind duration exceeds t , the sea is fetch-limited and these values apply directly.

4. Duration-limited and fully-developed limits

If the wind blows for a shorter time t_d , the duration relation is inverted to an **equivalent fetch**, and the height and period are recomputed from it (they come out smaller):

$$X_{eq} = (g \cdot t_d / (68.8 \cdot U_A))^3, \quad F_{eq} = X_{eq} \cdot U_A^2 / g; \quad \text{use the smaller of } F \text{ and } F_{eq}.$$

Finally the result is capped at the fully-developed limits for the wind speed: $gH_{m0,max}/U_A^2 = 0.2433$, $gT_{p,max}/U_A = 8.134$.

5. Growth curves

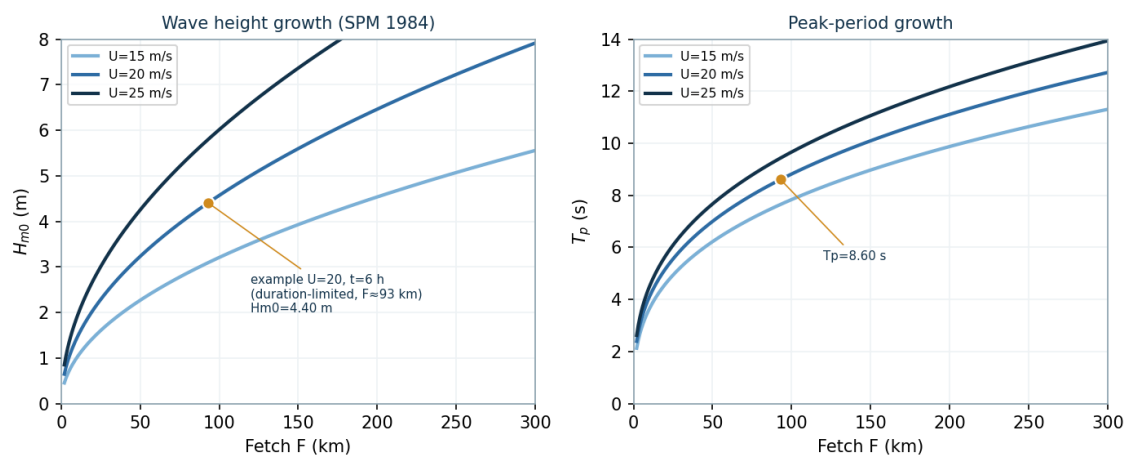


Figure 1. Significant wave height (left) and peak period (right) versus fetch for three wind speeds, after SPM (1984). Height grows as $F^{1/2}$, period as $F^{1/3}$. The marker is the worked example, reduced by duration-limiting to an equivalent fetch of ~93 km.

6. Worked example

Quantity	Value
Inputs	$U = 20 \text{ m/s}$, $F = 100 \text{ km}$, $t = 6 \text{ h}$
Adjusted wind, $U_A = 0.71 \cdot U^{1.23}$	28.3 m/s
Dimensionless fetch, $X = gF/U_A^2$	1226

Fetch-limited H_{m0} / T_p	4.57 m / 8.82 s
Duration needed for fetch-limit	6.3 h
Duration check	$t = 6 \text{ h} < 6.3 \text{ h} \rightarrow \text{duration-limited}$
Equivalent fetch, F_{eq}	92.8 km ($\approx 93 \text{ km}$)
Governing H_{m0} / T_p	4.40 m / 8.60 s
Fully-developed limits (reference)	19.8 m / 23.5 s

Notation

U wind speed at 10 m; U_A adjusted wind (wind-stress factor); F fetch; t / t_d wind duration; X dimensionless fetch; H_{m0} significant (spectral) wave height; T_p peak period; F_{eq} equivalent (duration) fetch; g gravitational acceleration. Estimates are deep-water and open-coast; local sheltering, shallow-water and gustiness effects are not included.

References

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